Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. $(x^3 + y^2 + 2)\,dx + 2y\,dy = 0$

Ans. $y^2 = 3x^2 - 6x - x^3 + ce^{-x} + 4$

Sol. $(x^3 + y^2 + 2)dx + 2ydy = 0$ _____(1)

Let $y^2 + 2 = v$

$\Rightarrow 2ydy = dv$

Substituting values in equation (1)

$\Rightarrow (x^3 + v)dx + dv = 0$

$$\Rightarrow \frac{dv}{dx} + v = -x^3$$

$$\text{I.F.} = e^{\int 1dx} = e^x$$

$$\Rightarrow v \cdot e^x = \int -x^3 e^x dx + C$$

$$\Rightarrow v \cdot e^x = -\left(x^3 - 3x^2 + 6x - 6\right)e^x + C$$

$$\Rightarrow v = -x^3 + 3x^2 - 6x + 6 + ce^{-x}$$

$$y^2 = -x^3 + 3x^2 - 6x + 4 + ce^{-x}$$

2. $$\left(\frac{dy}{dx}\right)^2 - (x+y)\frac{dy}{dx} + xy = 0$$

Ans. $y = ce^x$; $y = c + \frac{x^2}{2}$

Sol. $$\left(\frac{dy}{dx} - y\right)\left(\frac{dy}{dx} - x\right) = 0$$

$$\frac{dy}{dx} = y \quad \text{or} \quad \frac{dy}{dx} = x$$

$$\Rightarrow \ell ny = x + c \quad \text{or} \quad y = \frac{x^2}{2} + c$$

$$\Rightarrow y = ke^x$$

3. $\frac{dy}{dx} = \frac{y^2 - x}{2y\,(x+1)}$

Ans. $y^2 = -1 + (x+1)\ \ell n\ \frac{c}{x+1}$

Sol. $\frac{dy}{dx} = \frac{y^2 - x}{2y(x+1)}$

$\Rightarrow 2y\frac{dy}{dx} - \frac{y^2}{x+1} = \frac{-x}{x+1}$

putting $y^2 = t$

$\Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} - \frac{t}{x+1} = \frac{-x}{1+x}$

I.F. $= e^{-\int \frac{1}{x+1}dx} = e^{\ell n \frac{1}{x+1}} = \frac{1}{x+1}$

$\frac{t}{x+1} = \int \frac{1}{(1+x)}\cdot\left(\frac{-x}{1+x}\right)dx$

$= \int\left(\frac{1}{(1+x)^2} - \frac{1}{(1+x)}\right)dx$

$\frac{t}{1+x} = \frac{-1}{(1+x)} - \ell n(1+x) + \ell nC$

$t = -1 + (1+x)\,\ell n\left(\frac{C}{1+x}\right)$

$y^2 = -1 + (1+x)\,\ell n\left(\frac{C}{1+x}\right)$

4. $(1 - xy + x^2y^2)\,dx = x^2\,dy$

Ans. $y = \frac{1}{x}\tan\left(\ell n\,|cx|\right)$

Sol. $xy = t \Rightarrow x\frac{dy}{dx} + y = \frac{dt}{dx}$

$\Rightarrow x\frac{dy}{dx} = \frac{dt}{dx} - \frac{t}{x}$

$\Rightarrow (1 - t + t^2) = x\left(\frac{dt}{dx} - \frac{t}{x}\right)$

$\Rightarrow \frac{(1 - t + t^2)}{x} + \frac{t}{x} = \frac{dt}{dx} \Rightarrow \frac{1 + t^2}{x} = \frac{dt}{dx}$

$\Rightarrow \frac{dt}{1 + t^2} = \frac{dx}{x} \quad \Rightarrow \quad \tan^{-1}t = \ell n|x| + \ell nc$

$\Rightarrow \tan^{-1}(xy) = \ell n|cx|$

$xy = \tan \ell n|xc|$

5. $(x^2 + y)\frac{dy}{dx} = 6x$

Ans. $y = 3\ ln(x^2 + y + 3) + C$

Sol. $\left(x^2 + y\right)\frac{dy}{dx} = 6x$

$\Rightarrow 6x\frac{dx}{dy} = \left(x^2 + y\right)$

Let $x^2 = t \Rightarrow 2x\frac{dx}{dy} = \frac{dt}{dy}$

$\Rightarrow 3\frac{dt}{dy} = \left(t + y\right)$

$\Rightarrow \frac{dt}{dy} - \frac{t}{3} = \frac{y}{3}$

$\text{I.F.} = e^{\int -\frac{1}{3}dy} = e^{-\frac{y}{3}}$

$t \cdot e^{-\frac{y}{3}} = \int e^{-\frac{y}{3}}\frac{y}{3}dy = -\left(y + 3\right)e^{-\frac{y}{3}} + C$

$\Rightarrow x^2 = -\left(y + 3\right) + Ce^{\frac{y}{3}}$

6. $\frac{dy}{dx} - y\,ln2 = 2^{\sin x}.(\cos x - 1)\,ln2$, y being bounded when $x \to +\infty$.

$\frac{dy}{dx} - y\,ln2 = 2^{\sin x}.(\cos x - 1)\,ln2$, $x \to +\infty$ के लिए y परिबद्ध है

Ans. $y = 2^{\sin x}$

Sol. $\frac{dy}{dx} - y\log_e 2 = 2^{\sin x}(\cos x - 1)\log_e 2$

This is linear differential equation.

$\text{I.F.} = e^{-\log_e 2\int dx} = e^{-x\log_e 2} = 2^{-x}$

Solution is

$y2^{-x} = \int 2^{-x}2^{\sin x}(\cos x - 1)\log_e 2dx$

put $\sin x - x = t \Rightarrow (\cos x - 1)dx = dt$

$\therefore y2^{-x} = \log_e 2\int 2^t dt$

$\therefore\ y2^{-x} = 2^t + c$

$\therefore\ y = 2^{x+t} + c2^x$

$\therefore\ y = 2^{\sin x} + c2^x$

7. Find the curve which passes through the point (2, 0) such that the segment of the tangent between the point of tangency and the y–axis has a constant length equal to 2.

बिन्दु वक्र (2, 0) से गुजरने वाला वक्र ज्ञात कीजिए जिसके किसी स्पर्शी का स्पर्श बिन्दु एवं y–अक्ष के मध्य अन्तःखण्ड की लम्बाई 2 हों।

Ans. $y = \pm\left[\sqrt{4 - x^2} + 2\ell n\frac{2 - \sqrt{4 - x^2}}{x}\right]$

Sol. Let the curve be $y = f(x)$ and slope of tangent at any point $P(x, y) = \frac{dy}{dx}$

So, the equation of the tangent :

$Y - y = \frac{dy}{dx}(X - x)$

Let tangent meet y-axis at point N and y-axis, put $X = 0$.

$Y = y - x\frac{dy}{dx}$

So, the coordinates of the point $N = \left(0, y - x\frac{dy}{dx}\right)$

Given that PN = 2.

$$\Rightarrow \sqrt{(0-x)^2 + \left(y - x\frac{dy}{dx} - y\right)^2} = 2$$

$$\Rightarrow x^2 + x^2\left(\frac{dy}{dx}\right)^2 = 4$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{4-x^2}{x^2}$$

$$\Rightarrow \frac{dy}{dx} = \pm\sqrt{\frac{4-x^2}{x^2}}$$

$$\Rightarrow dy = \pm\left(\sqrt{\frac{4-x^2}{x^2}}\right)dx$$

On integrating both sides and let $x = 2\sin\theta$ _____(1)

$$\Rightarrow dx = 2\cos\theta d\theta$$

$$\Rightarrow \int dy = \pm 2\int \frac{\cos^2\theta}{\sin\theta} d\theta$$

$$\Rightarrow \int dy = \pm 2\int \left(\frac{1-\sin^2\theta}{\sin\theta}\right) d\theta$$

$$\Rightarrow \int dy = \pm 2\int (\operatorname{cosec}\theta - \sin\theta) d\theta$$

$$\Rightarrow y = \pm 2\left[\ln(\operatorname{cosec}\theta - \cos\theta) - (-\cos\theta)\right] + c$$ _____(2)

Since from equation (1)

$$\sin\theta = \frac{x}{2}$$

$$\Rightarrow \cos\theta = \sqrt{1-\frac{x^2}{4}} = \frac{1}{2}\sqrt{4-x^2}$$

$$\cos ec\theta = \frac{2}{x}$$

$$\cot\theta = \frac{\sqrt{4-x^2}}{x}$$

Putting all these values in equation (2), we get

$$y = \pm 2\left[\ln\left(\frac{2}{x} - \frac{\sqrt{4-x^2}}{x}\right) + \frac{\sqrt{4-x^2}}{2}\right] + c$$

Since, the curve passes through the point (2,0).

$\Rightarrow c = 0$

Therefore, $y = \pm\left[\sqrt{4-x^2} + 2\ln\frac{2-\sqrt{4-x^2}}{x}\right]$

8. $x\,dy + y\,dx + \frac{x\,dy - y\,dx}{x^2+y^2} = 0$

Ans. $xy + \tan^{-1}\frac{y}{x} = c$

Sol. $xdy + ydx + \frac{xdy - ydx}{x^2+y^2} = 0$

$$\Rightarrow d(xy) + d\tan^{-1}\left(\frac{y}{x}\right) = 0$$

$$\Rightarrow xy + \tan^{-1}\frac{y}{x} = c$$

9. $\frac{y\,dx - x\,dy}{(x-y)^2} = \frac{dx}{2\sqrt{1-x^2}}$, given that y=2 when x = 1.

Ans. $\frac{\sin^{-1}x}{2} - \frac{y}{x-y} = \frac{\pi}{4} + 2$

Sol. $y - \frac{xdy}{dx} = \frac{(x-y)^2}{2\sqrt{1-x^2}}$

$$\Rightarrow (y-x)+x-\frac{xdy}{dx}=\frac{(x-y)^2}{2\sqrt{1-x^2}}$$

$$\Rightarrow \frac{x}{(x-y)^2}\left(1-\frac{dy}{dx}\right)-\frac{1}{(x-y)}=-\frac{1}{2\sqrt{1-x^2}}$$

$$\Rightarrow \frac{1}{(x-y)^2}\left(1-\frac{dy}{dx}\right)-\frac{1}{x(x-y)}=-\frac{1}{2x\sqrt{1-x^2}}$$

Let $-\frac{1}{x-y}=t \Rightarrow \frac{1}{(x-y)^2}\left(1-\frac{dy}{dx}\right)=\frac{dt}{dx}$

$$\Rightarrow \frac{dt}{dx}+\frac{t}{x}=-\frac{1}{2x\sqrt{1-x^2}}$$

$$\text{I.F.}=e^{\int\frac{1}{x}dx}=x$$

$$t\cdot x=-\int\frac{x}{2x\sqrt{1-x^2}}dx$$

$$\Rightarrow t\cdot x=-\frac{1}{2}\int\frac{1}{\sqrt{1-x^2}}dx$$

$$\Rightarrow t\cdot x=-\frac{1}{2}\sin^{-1}x+c$$

$$\Rightarrow -\frac{x}{x-y}=-\frac{1}{2}\sin^{-1}x+c$$

or

$$\Rightarrow -\frac{y}{x-y}=-\frac{1}{2}\sin^{-1}x+c^1$$

Now, (x, y) = (1,2)

$$\Rightarrow \frac{-2}{-1}=-\frac{1}{2}\sin^{-1}1+c^1 \Rightarrow c^1=2+\frac{\pi}{4}$$

$$\frac{\sin^{-1}x}{2}-\frac{y}{x-y}=\frac{\pi}{4}+2$$

10. Find the equation of the curve passing through the orgin if the middle point of the segment of its normal from any point of the curve to the x-axis lies on the parabola $2y^2 = x$.

मूल बिन्दु से गुजरने वाले उस वक्र का समीकरण ज्ञात कीजिए। यदि वक्र के किसी बिन्दु पर खीचें गये लम्ब के बिन्दु एवं x-अक्ष के मध्य भाग का मध्य बिन्दु परवलय $2y^2 = x$ पर पड़ता है।

Ans. $y^2 = 2x + 1 - e^{2x}$

Sol. Equation of normal at any point P(x, y), is

$$\frac{dy}{dx}(Y-y)+(X-x)=0$$

This meets the x-axis at $A\left(x+y\frac{dy}{dx},0\right)$

Midpoint of AP is $\left(x+\frac{1}{2}y\frac{dy}{dx},\frac{y}{2}\right)$ which lies on the parabola $2y^2 = x$.

Therefore, $2\times\frac{y^2}{4}=x+\frac{1}{2}y\frac{dy}{dx}$ or $y^2=2x+y\frac{dy}{dx}$

Put $y^2 = t$ $\Rightarrow 2y\frac{dy}{dx}=\frac{dt}{dx}$

we get $\frac{dt}{dx}-2t=-4x$ (linear)

$I.F.e^{-2\int(dx)}=e^{-2x}$

Therefore, solution is given by

$$te^{-2x}=-4\int xe^{-2x}dx+c$$

$$=-4\left[-\frac{1}{2}xe^{-2x}-\int-\frac{1}{2}e^{-2x}1dx\right]+c$$

$$y^2e^{-2x}=2xe^{-2x}+e^{-2x}+c$$

Since, the curve passes through (0,0), c = –1. Therefore,

$y^2 = 2x + 1 - e^{2x}$ is the equation of the required curve.

11. Find the continuous function which satisfies the relation, $\int_0^x t\, f(x-t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$, for all real number x.

वह सतत् फलन ज्ञात कीजिए जो संबंध $\int_0^x t\, f(x-t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$, को x के सभी वास्तविक मानों के लिये संतुष्ट करता है।

Ans. $f(x) = e^x - \cos x$

Sol. Using $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

$$\Rightarrow \int_0^x tf(x-t)\,dt = \int_0^x (x-t) f(t)\,dt$$

$$\text{Now, } \int_0^x (x-t) f(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$$

$$\Rightarrow x\int_0^x f(t)\,dt - \int_0^x tf(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$$

Differentiating both sides

$$\Rightarrow \int_0^x f(t)\,dt + xf(x) - xf(x) = f(x) + \cos x - \sin x - 1$$

$$\Rightarrow \int_0^x f(t)\,dt = f(x) + \cos x - \sin x - 1 \qquad ...(1)$$

Again differentiating both sides

$$\Rightarrow f(x) = f'(x) - \cos x - \sin x \qquad (y = f(x))$$

$$\Rightarrow \frac{dy}{dx} - y = \sin x + \cos x$$

$$IF = e^{\int -1dx} = e^{-x}$$

$$\Rightarrow ye^{-x} = \int e^{-x}(\sin x + \cos x)\,dx \qquad ...(2)$$

$\int e^{-x}(\sin x + \cos x)dx \rightarrow$ Put $x = -t$ & $dx = -dt$

$\Rightarrow -\int e^{t}(\cos t - \sin t)dt = -e^{t}\cos t \quad \left[\int e^{x}(f(x)) + f'(x)dx = e^{x}f(x)\right]$

Put in equation (2)

$ye^{-x} = -e^{-x}\cos x + c$

$y = ce^{x} - \cos x$

Put $x = 0$ in equation (1), $f(0) = 0$

$y = ce^{x} - \cos x$

Put $(x, y) = (0,0) \Rightarrow 0 = c - 1 \Rightarrow c = 1$

$y = e^{x} - \cos x$

12. Given two curves $y = f(x)$ passing through the point (0, 1) & $y = \int_{-\infty}^{x} f(t)dt$ passing through the point (0, 1/2). The tangents drawn to both curves at the points with equal abscissas intersect on the x–axis. Find the curve f(x).

वक्र $y = f(x)$ बिन्दु (0, 1) से तथा वक्र $y = \int_{-\infty}^{x} f(t)dt$, (0, 1/2) से गुजरते है। दोनों वक्रो पर समान भुज वाले बिन्दुओं पर खीचीं स्पर्श रेखाएं x अक्ष पर मिलती है। वक्र f(x) ज्ञात कीजिए।

Ans. $f(x) = e^{2x}$

Sol. Equation of tangent to the curve $y = f(x)$ is

$Y - y = f'(x)(X - x)$

Equation of tangents to the curve $g(x) = y_1 = \int_{-\infty}^{x} f(t)dt$ is $Y - y = f(x)(X - x)\left(\frac{dy_1}{dx} = g'(x) = f(x)\right)$

Since the tangent with equal abscissa intersect on the x-axis

$\Rightarrow x - \frac{y}{f'(x)} = x - \frac{y_1}{f(x)}$

$\Rightarrow \frac{f(x)}{f'(x)} = \frac{y_1}{f(x)}$

$\Rightarrow \frac{g'(x)}{g(x)} = \frac{g''(x)}{g'(x)}$

$\Rightarrow \ell n\, g(x) = \ln cg'(x)$

$\Rightarrow g(x) = cg'(x)$

$$\Rightarrow \frac{g'(x)}{g(x)} = c$$

$\Rightarrow g(x) = ke^{cx}$

$\Rightarrow f(x) = g'(x) = kce^{cx}$

The curve $y = f(x)$ passes through $(0,1) \Rightarrow kc = 1$

$g(x) = ke^{cx}$ passing through $\left(0, \frac{1}{2}\right)$

$$\Rightarrow \frac{1}{2} = k \Rightarrow kc = 1 \Rightarrow c = 2$$

Hence, $f(x) = e^{2x}$

13. Find the integral curve of the differential equation, $x(1 - x\, l n y).\frac{dy}{dx} + y = 0$ which passes through $\left(1, \frac{1}{e}\right)$.

अवकल समीकरण $x(1 - x\, l n y).\frac{dy}{dx} + y = 0$ का समाकल वक्र ज्ञात कीजिए जो $\left(1, \frac{1}{e}\right)$ से गुजरता है।

Ans. $x(ey + l n\, y + 1) = 1$

Sol. $$x\left(1 - x\ell ny\right)\frac{dy}{dx} + y = 0$$

$$\Rightarrow y\frac{dx}{dy} + x = x^2 \ell ny \qquad \text{_____(1)}$$

Let $\ell n\, y = z$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow y\frac{dx}{dy} = \frac{dx}{dz}$$

$$\Rightarrow \frac{dx}{dz} + x = x^2 z$$

$$\Rightarrow \frac{1}{x^2}\frac{dx}{dz}+\frac{1}{x}=z$$

which is a linear differential equation.

Let $\frac{1}{x}=v$

$$\Rightarrow -\frac{1}{x^2}\frac{dx}{dz}=\frac{dv}{dz}$$

$$\Rightarrow -\frac{dv}{dz}+v=z$$

$$\Rightarrow \frac{dv}{dz}-v=-z$$

which is a linear differential equation.

$\therefore IF = e^{-\int dz} = e^{-z}$

Hence the solution is

$$v.e^{-z}=\int -ze^{-z}dz+c$$

$$\Rightarrow v\cdot e^{-z}=-\left(-z-1\right)e^{-z}+c$$

$$\Rightarrow v=\left(z+1\right)+ce^{z}$$

$$\Rightarrow \frac{1}{x}=\left(\ell ny+1\right)+ce^{\ell ny}$$

$$\Rightarrow \frac{1}{x}=\left(\ell ny+1\right)+cy$$

which is passing through $\left(e,\frac{1}{e}\right)$.

So, c = e

Hence, the required integral curves is $\frac{1}{x}=\left(\ell ny+1\right)+ey$

14. Find all the curves possessing the following property; the segment of the tangent between the point of tangency & the x-axis is bisected at the point of intersection with the y-axis.

वह सभी वक्र ज्ञात कीजिए जिसके लिये स्पर्श बिन्दु एवं x अक्ष के बीच स्पर्श रेखा का अन्तःखण्ड, y-अक्ष पर समद्विभाजित होता है।

Ans. $y^2 = cx$

Sol. Equation of tangent at P(x, y) is

$$Y - y = \frac{dy}{dx}(X - x)$$

It meets x-axis at Q(x, 0)

$$-y = \frac{dy}{dx}(X - x)$$

$$x - y\frac{dx}{dy} = X$$

$$Q \equiv \left(x - y\frac{dx}{dy}, 0\right)$$

PQ bisected at point of intersection at y-axis let R (0,y)

$$\Rightarrow \frac{x + x - y\frac{dx}{dy}}{2} = 0$$

$$2x = y\frac{dx}{dy}$$

$$\int \frac{dy}{y} = \int \frac{dx}{2x}$$

$$\log y = \frac{1}{2}\log x + \log c$$

$y^2 = cx$

15. A curve passing through (1 , 0) such that the ratio of the square of the intercept cut by any tangent off the y-axis to the subnormal is equal to the ratio of the product of the co-ordinates of the point of tangency to the product of square of the slope of the tangent and the subtangent at the same point. Determine all such possible curves.

एक वक्र जो (1 , 0) से गुजरता है, के स्पर्श रेखा द्वारा y-अ़क्ष पर अन्तःखण्ड की लम्बाई का वर्ग एवं अर्धलम्ब का अनुपात, स्पर्श बिन्दु के निर्देशांको के गुणनफल एवं उसी बिन्दु पर स्पर्श रेखा के ढाल का वर्ग तथा अधःस्पर्शी के गुणनफल के अनुपात के बराबर है। ऐसे सभी वक्रो को ज्ञात कीजिए।

Ans. $x = e^{2\sqrt{y/x}}; e^{-2\sqrt{y/x}}$

Sol. Tangent at P(x, y)

$$\Rightarrow Y - y = \frac{dy}{dx}(X - x) \text{ for y intercept } x = 0.$$

$$\Rightarrow Y = y - x\frac{dy}{dx}$$

Given $\frac{Y^2}{my} = \frac{xy}{m^2\frac{y}{m}}$

$$\Rightarrow \left(y - x\frac{dy}{dx}\right)^2 = xy$$

$$\Rightarrow \left(\frac{y}{x} - \frac{dy}{dx}\right)^2 = \frac{y}{x}$$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\Rightarrow \left(v - v - x\frac{dv}{dx}\right)^2 = v$$

$$\Rightarrow \left|x\frac{dv}{dx}\right| = \sqrt{v} \Rightarrow \int \frac{dv}{\pm\sqrt{v}} = \int \frac{dx}{x}$$

$$\Rightarrow \pm 2\sqrt{v} = \ell nx + \ell nc$$

$$\Rightarrow xc = e^{\pm 2\sqrt{\frac{y}{x}}} \text{ passing from } (1,0)$$

$$\Rightarrow c = 1 \Rightarrow x = e^{\pm 2\sqrt{\frac{y}{x}}}$$

16. A & B are two separate reservoirs of water. Capacity of reservoir A is double the capacity of reservoir B. Both the reservoirs are filled completely with water, their inlets are closed and then the water is released simultaneously from both the reservoirs. The rate of flow of water out of each reservoir at any instant of time is proportional to the quantity of water in the reservoir at that time. One hour after the water is released, the quantity of water in reservoir A is 1.5 times the quantity of water in reservoir B. After how many hours do both the reservoirs have the same quantity of water ?

A एवं B दो भिन्न पानी की टंकिया है। A की क्षमता B की क्षमता से दोगुनी है। दोनों टंकियों को पूरी तरह पानी से भरकर उनके नलों को बंद करके दोनों टंकियों से एक साथ पानी छोड़ा जाता है। किसी टंकी में पानी के बहाव की दर, उस समय टंकी में पानी की मात्रा के समानुपातिक है। पानी खोलने के 1 घण्टे पश्चात् A टंकी में पानी B की 1.5 गुना होता है। कितने घंटे पश्चात् दोनों टंकियों में समान मात्रा में पानी होगा ?

Ans. $T = \log_{4/3} 2$ hrs from the start

Sol. Let at time t hours, the volume of water in A and B be u and v, respectively. From the question,

$$\frac{du}{dt} = -k_1 u \qquad \text{_____(1)}$$

and $$\frac{dv}{dt} = -k_2 v \qquad \text{_____(2)}$$

At $t = 0$, $u = 2v$ and $v = V$ (from the question). Solving Equation (1),

$$\frac{du}{u} = -k_1 dt$$

Integrating both sides, we get

$$\log u = -k_1 t + c_1$$

$$u = e^{-k_1 t + c_1} \qquad \text{_____(3)}$$

When $t = 0$, $u = 2v$. Then

$$2V = e^{c_1}$$

Hence, equation (3) gives

$$u = 2Ve^{-k_1 t} \qquad \text{_____(4)}$$

Similarly, from equation (2), we get

$$v = Ve^{-k_2 t} \qquad \text{_____(5)}$$

From the question, if $v = v_0$ when $t = 1$, then $u = \frac{3}{2} v_0$

Therefore, equation (4) gives

$$\frac{3}{2} v_0 = 2Ve^{-k},$$

and equation (5) gives $v_0 = Ve^{-k_1}$

Dividing these, we get

$$\frac{3}{2}=\frac{2e^{-k_1}}{e^{-k_2}}=2e^{k_2-k_1}$$

Thus,

$$e^{k_2-k_1}=\frac{3}{4} \qquad \text{_____(6)}$$

Let after T hours, the volume of water in A and B be equal. Therefore, equations (4) and (5) gives

$$2Ve^{-k_1T}=Ve^{-k_2T}$$

$$\Rightarrow 2e^{-k_1T}=e^{-k_2T}\Rightarrow 2e^{(k_2-k_1)T}=1$$

Using equation (6), we get

$$2\cdot\left(\frac{3}{4}\right)^{T}=1$$

$$\Rightarrow\left(\frac{4}{3}\right)^{T}=2$$

Therefore,

$$T\log\frac{4}{3}=\log 2\Rightarrow T=\frac{\log 2}{\log\frac{4}{3}}=\log_{\frac{4}{3}}2$$

Hence, after $\log_{\frac{4}{3}}2$ hours, the reservoirs will have the same quantity of water.

17. A tank consists of 50 litres of fresh water. Two litres of brine each litre containing 5 gms of dissolved salt are run into tank per minute; the mixture is kept uniform by stirring, and runs out at the rate of one litre per minute. If 'm' grams of salt are present in the tank after t minute, express 'm' in terms of t and find the amount of salt present after 10 minutes.

किसी टंकी में 50 लीटर ताजा पानी है। एक लीटर में 5 gms घुला लवण वाला 2 लीटर ब्राइन, प्रति मिनट टंकी में जाता है। मिश्रण को हिलाकर एक समान बनाया जाता है, तथा 1 लीटर प्रति मिनट की दर से बाहर निकलता है। यदि t समय पश्चात् टंकी में 'm' ग्राम लवण उपस्थित हो तो 'm' को t के पदों में व्यक्त करो तथा 10 मिनट पश्चात् उपस्थित लवण की मात्रा बताओं।

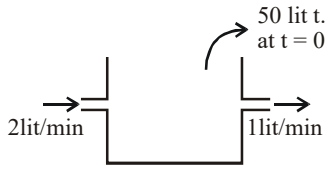
Ans. $y=5t\left(1+\frac{50}{50+t}\right)\text{gms}\,;\,91\frac{2}{3}\text{ gms}$

Sol. Input rate = 10 gm/min

After time t volume of tank = 50 + (2 – 1)t

Concentration of salt in time $t = \frac{m}{50+t}$ gm/lit

Output rate $= \frac{m}{50+t}.1$ gm/min.



$$\frac{dm}{dt} = \frac{-m}{50+t} + 10$$

$$m(50+t) = \int 10(50+t)\,dt$$

$\Rightarrow$ $m(50+t) = 500\,t + 5t^2 + c$

$\Rightarrow$ ($\because$ at $t = 0$, $m = 0 \Rightarrow c = 0$)

$\Rightarrow$ $m(50+t) = 5\,(100\,t + t^2)$

$\Rightarrow$ $m = 5t\left(\frac{100+t}{50+t}\right)$ gm

18. Find the curve for which the portion of y-axis cut-off between the origin and the tangent varies as cube of the absissa of the point of contact.

किसी वक्र पर स्पर्शी एवं मूल बिन्दु के बीच y-अक्ष का कटा अन्तःखण्ड, स्पर्श बिन्दु के भुज के घन के समानुपातिक है। वक्र का समीकरण बताओं।

Ans. $2y + Kx^3 = cx$

Sol. Equation of tangent at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x), \text{ for y-axis, } X = 0$$

$$\Rightarrow Y = y - x\frac{dy}{dx}$$

Given $y - x\frac{dy}{dx} \propto x^3 \Rightarrow y - x\frac{dy}{dx} = kx^3$

$$\therefore \frac{dy}{dx} - \frac{y}{x} = kx^2 \qquad \left\{IF = \frac{1}{x}\right\}$$

$$\Rightarrow y = \frac{-kx^3}{2} + cx$$

19. Find the equation of the curve which passes through the origin and the tangent to which at every point (x, y) has slope equal to $\frac{x^4 + 2xy - 1}{1 + x^2}$.

मूल बिन्दु से गुजरने वाले उस वक्र का समीकरण ज्ञात कीजिए, जिसके किसी बिन्दु (x, y) पर स्पर्शी की प्रवणता $\frac{x^4+2xy-1}{1+x^2}$ हो।

Ans. $y=(x-2\tan^{-1}x)(1+x^2)$

Sol. It is given that

$$\frac{dy}{dx}=\frac{x^4+2xy-1}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx}-\frac{2x}{1+x^2}y=\frac{x^4-1}{1+x^2}$$

which is a linear differential equation.

$$\therefore IF=e^{-\int\frac{2x}{1+x^2}dx}=e^{-\log\left(x^2+1\right)}=\frac{1}{\left(x^2+1\right)}$$

Hence, the solution is

$$y\cdot\frac{1}{\left(x^2+1\right)}=\int\frac{x^4-1}{\left(x^2+1\right)^2}dx+c$$

$$\Rightarrow \frac{y}{\left(x^2+1\right)}=\int\frac{x^2-1}{\left(x^2+1\right)}dx+c$$

$$=\int\left(1-\frac{2}{x^2+1}\right)dx+c$$

$=(x-2\tan^{-1}x)+c$

20. A right circular cone with radius R and height H contains a liquid which evaporates at a rate proportional to its surface area in contact with air (proportionality constant = k > 0). Find the time after which the cone is empty.

৮ ऊँचाई और R त्रिज्या का एक प्रतिलोम शंकु पैंदे पर रखा हुआ है। इसे एक वाष्पशील द्रव से पूरा भरा जाता है। यदि वाष्पन की दर हवा के सम्पर्क में रहने वाले द्रव के पृष्ठीय क्षेत्रफल के समानुपाती है (समानुपाती नियतांक = k > 0 है), तो वह समय ज्ञात कीजिए जिसमें पूरा द्रव वाष्पित हो जायेगा।

Ans. T = H/k

Sol. Given, liquid evaporates at a rate proportional to its surface area.

$$\Rightarrow \frac{dV}{dt}\propto -S \qquad \text{_____(1)}$$

We know, volume of cone $= \frac{1}{3}\pi r^2 h$

and surface area $= \pi r^2$

or $V = \frac{1}{3}\pi r^2 h$ and $S = \pi r^2$ _____(2)

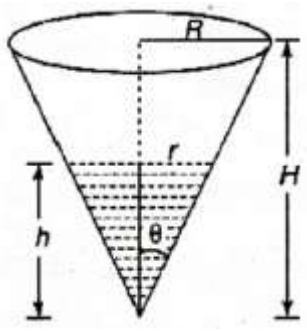
where $\tan\theta = \frac{R}{H}$ and $\frac{r}{h} = \tan\theta$ _____(3)

From equations (ii) and (iii), we get

$V = \frac{1}{3}\pi r^3 \cot\theta$ and $S = \pi r^2$ _____(4)

On substituting equation (4) in equation (1), we get

$$\frac{\pi}{3}\cot\theta \cdot 3r^2 \frac{dr}{dt} = -k\pi r^2$$



$$\Rightarrow \cot\theta \int_R^0 dr = -k\int_0^T dt$$

$$\Rightarrow \cot\theta(0 - R) = -k(T - 0)$$

$$\Rightarrow R\cot\theta = kT$$

$\Rightarrow H = kT$ [From equation (3)]

$$\Rightarrow T = \frac{H}{k}$$

21. If $P(1) = 0$ and $\frac{dP(x)}{dx} > P(x)$ for all $x \geq 1$ then prove that $P(x) > 0$ for all $x > 1$.

यदि $P(1) = 0$ तथा $\frac{dP(x)}{dx} > P(x)\ \ \forall\ x \geq 1$ हो, तो सिद्ध कीजिये कि $P(x) > 0\ \ \forall\ x > 1$.

Ans.

Sol. Given, $P(1) = 0$ and

$$\frac{dP(x)}{dx} - P(x) > 0, \forall x \geq 1 \qquad ...(1)$$

On multiplying by e^{-x} ($e^{-x} > 0$)

$$\Rightarrow e^{-x} \cdot \frac{d}{dx} P(x) + P(x) \cdot \frac{d}{dx} e^{-x} > 0$$

$$\Rightarrow \frac{d}{dx}\left(P(x) \cdot e^{-x}\right) > 0$$

$\Rightarrow P(x) \cdot e^{-x}$ is an increasing function

$$\Rightarrow P(x) \cdot e^{-x} > P(1) \cdot e^{-1}, \forall x \geq 1$$

$$\Rightarrow P(x) > 0, \forall x > 1 \ [P(1) = 0 \text{ and } e^{-x} > 0]$$

22. Let f(x) be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that $\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$. Find the value of $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$.

माना f(x) अन्तराल $[-1, \infty)$ में अवकलनीय फलन है तथा $f(0) = 1$ इस प्रकार है कि $\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$ है, तो $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$ का मान ज्ञात कीजिए।

Ans. 1

Sol. $h = t - (x + 1)$

$$= \lim_{h \to 0} \frac{\left(h + (x+1)\right)^2 f(x+1) - (x+1)^2 f\left(h + (x+1)\right)}{f\left(h + (x+1)\right) + f(x+1)}$$

$$= \lim_{h \to 0} \frac{\left(h^2 + 2h(x+1)\right) f(x+1) - \left[f\left(h + (x+1)\right) - f(x+1)\right](x+1)^2}{f\left(h + (x+1)\right) - f(x+1)}$$

$$= \lim_{h\to 0} \frac{\big(h+2(x+1)\big)f(x+1) - \dfrac{\big(f(h+(x+1)) - f(x+1)\big)(x+1)^2}{h}}{\dfrac{\big(f(h+(x+1)) - f(x+1)\big)}{h}}$$

$$= \frac{2(x+1)f(x+1) - f'(x+1)(x+1)^2}{f'(x+1)}$$

Now, $\dfrac{2(x+1)f(x+1) - f'(x+1)(x+1)^2}{f'(x+1)} = 1$

$$\Rightarrow f'(x+1)\Big((x+1)^2+1\Big) - 2(x+1)f(x+1) = 0$$

Replace $x + 1$ by x

$$\Rightarrow f'(x)\left(x^2+1\right) - 2xf(x) = 0$$

$$\Rightarrow \frac{dy}{dx} - \frac{2xy}{x^2+1} = 0 \Rightarrow \frac{dy}{y} - \frac{2x}{x^2+1}dx = 0$$

$\Rightarrow \ell n\, y - \ell n\,(x^2 + 1) = \ell n\, c$

$\Rightarrow y = c(x^2 + 1)$ at $x = 0, y = 1$

$\Rightarrow c = 1 \quad \Rightarrow y = (x^2 + 1)$

Now, $\lim\limits_{x\to 1} \dfrac{\ell n f(x) - \ell n 2}{x-1}$

$$= \lim_{x\to 1} \frac{\ell n \dfrac{x^2+1}{2}}{x-1} = \lim_{x\to 1} \frac{\dfrac{x^2+1}{2} - 1}{x-1}$$

$$= \frac{1}{2}\lim_{x\to 1} \frac{x^2-1}{x-1} = \frac{1}{2}\lim_{x\to 1}(x+1) = 1$$

23. Let $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$, where a, b, c and d are parameters represent a family of curves, then differential equation for the given family of curves is given by $y'' - \alpha y' + \beta y = 0$, then $\alpha + \beta =$

माना $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$, जहाँ a, b, c तथा d प्राचल है, एक वक्र निकाय को निरूपित करता है, तो दिये गये वक्र fud k; d kv od y 1 ehdj .ky'' – $\alpha y' + \beta y = 0$ है, तो $\alpha + \beta =$

Ans. 4

Sol. $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$

$y = (A \sin x + B \cos x)\, e^{x}$ ...(1)

$y' = e^{x}(A \sin x + B \cos x) + e^{x} (A \cos x - B \sin x)$

$y' = y + e^{x} (A \cos x - B \sin x)$ ...(2)

$y'' = y' + e^{x} (A \cos x - B \sin x) + e^{x}(-A \sin x - B \cos x)$

$= y' + y' - y - y$

$\Rightarrow y'' - 2y' + 2y = 0, \quad \alpha = 2, \beta = 2$

$\Rightarrow \alpha + \beta = 4$

24. let $y = f(x)$ satisfies the differential equation $xy(1 + y)\, dx = dy$. If $f(0) = 1$ and $f(2) = \dfrac{e^2}{k - e^2}$, then find the value of k.

माना $y = f(x)$ अवकल समीकरण $xy(1 + y)\, dx = dy$ को संतुष्ट करता है। यदि $f(0) = 1$ तथा $f(2) = \dfrac{e^2}{k - e^2}$ है, तो k का मान होगा।

Ans. 2

Sol. $xy(1 + y)dx = dy$

$$\Rightarrow x\,dx = \frac{1}{y(1+y)}\,dy$$

$$\Rightarrow x\,dx = \frac{1+y-y}{y(1+y)}\,dy$$

$$\Rightarrow x\,dx = \frac{1}{y} - \frac{1}{1+y}\,dy$$

$$\Rightarrow \int x\,dx = \int \frac{1}{y} - \frac{1}{1+y}\,dy$$

$$\frac{x^2}{2} = \log y - \log(1+y) + \log C$$

$$f(0) = 1 \Rightarrow \frac{0^2}{2} = \log 1 - \log 2 + \log C \Rightarrow C = 2$$

Now, $x = 2 \Rightarrow \frac{4}{2} = \ell n f(2) - \ell n(1 + f(2)) + \ell n 2$

$\Rightarrow e^2 = \frac{2f(2)}{1+f(2)} \Rightarrow e^2 = \frac{-2}{1+f(2)}$

$\Rightarrow 1 + f(2) = \frac{2}{2-e^2}$

$\Rightarrow f(2) = \frac{e^2}{2-e^2}, k = 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 25 to 27

Given a function 'g' which has a derivative g′(x) for every real x and which satisfy g′(0) = 2 and $g(x + y) = e^y g(x) + e^x g(y)$ for all x and y.

दिया गया फलन 'g' का सभी वास्तविक x के लिए अवकलन g′(x) है तथा सभी x तथा y के लिए g′(0) = 2 तथा $g(x + y) = e^y g(x) + e^x g(y)$ को संतुष्ट करता है।

25. The function g(x) is :

फलन g(x) है :

(A) $x(2 + xe^x)$ (B) $x(e^x + 1)$ (C) $2xe^x$ (D) $x + \ln(x + 1)$

Ans. C

Sol. Put $x = 0; y = 0 \Rightarrow g(0) = 0$

and $g'(0) = \lim_{h\to 0} \frac{g(h)}{h} = 2$

Now, $g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$

$= \lim_{h\to 0} \frac{e^h g(x) + e^x g(h) - g(x)}{h}$

$= h(x) \lim_{h\to 0} \left(\frac{e^h - 1}{h}\right) + e^x \lim_{h\to 0} \frac{g(h)}{h}$

$\Rightarrow g'(x) = g(x) + 2e^x$

Let $g(x) = y$ then $\frac{dy}{dx} = y + 2e^x$

$\Rightarrow e^{-x}\frac{dy}{dx} - ye^{-x} = 2$

$\Rightarrow \frac{d}{dx}\left(ye^{-x}\right) = 2$

$\Rightarrow ye^{-x} = 2x + c$

Given if $x = 0, y = 0 \Rightarrow c = 0$

Then $ye^{-x} = 2x \Rightarrow y = 2xe^x$

26. The range of function g(x) is :

फलन g(x) का परिसर है :

(A) R (B) $\left[-\frac{2}{e}, \infty\right)$ (C) $\left[-\frac{1}{e}, \infty\right)$ (D) $[0, \infty)$

Ans. B

Sol. Now $\frac{dy}{dx} = 2\left[e^x + xe^x\right] = 0 \Rightarrow x = -1$

$\Rightarrow$ minima at $x = -1$

$\Rightarrow$ range is $\left[-\frac{2}{e}, \infty\right)$

27. The value of $\lim\limits_{x\to-\infty} g(x)$ is :

(A) 0 (B) 1 (C) 2 (D) Does not exist

$\lim\limits_{x\to-\infty} g(x)$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) विद्यमान नहीं है

Ans. A

Sol. $g(x) = 2xe^x$

$$\lim_{x\to-\infty} g(x) = 0$$

Paragraph for Question nos. 28 to 30

Let $f: R^+ \to R$ be a differentiable function satisfying $f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t)\,dt$ for all $x \in R^+$.

माना $f: R^+ \to R$ एक अवकलनीय फलन $f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t)\,dt$ सभी $x \in R^+$ के लिए संतुष्ट करता है।

28. The value of definite integral $\int_0^1 f(x)\,dx$ is equal to :

निश्चित समाकल $\int_0^1 f(x)\,dx$ का मान होगा :

(A) e (B) e – 2 (C) e + 1 (D) $\frac{e+1}{2}$

Ans. B

Sol. $f'(x) = 0 + (1-x)\left(\frac{1}{x}\right) + \ell n\left(\frac{x}{e}\right)(-1) + f(x)$ (By Newton's Leibnitz rule)

$$\Rightarrow f'(x) = \frac{1}{x} - 1 + (\ell nx - 1)(-1) + f(x)$$

$$\Rightarrow \frac{dy}{dx} - y = \frac{1}{x} - \ell nx$$

$$\Rightarrow ye^{-x} = \int\left(\frac{1}{x} - \ell nx\right)e^{-x}dx$$ (solving of linear differential equation)

$$\Rightarrow ye^{-x} = e^{-x}(\log x) + c$$

Since, $f(1) = e \Rightarrow ee^{-1} = 0 + c \Rightarrow c = 1$

$y = \log x + e^x$

$$\int_0^1 (\ell nx + e^x)\,dx = \left[x(\ell nx - 1) + e^x\right]_0^1$$

$= -1 + e - 0 - 1 = e - 2$

29. The x-intercept of normal drawn to the curve $y = f(x)$ at point P where $y = f(x)$ crosses the line $x = 1$, is equal to :

$y = f(x)$ के बिन्दु P जहाँ $y = f(x)$ रेखा $x = 1$ को काटता है, पर अभिलम्ब का x अंतःखण्ड होगा :

(A) $e^2 + e - 1$ (B) $\frac{e^2+e+1}{e+1}$ (C) $\frac{e^2-e+1}{e+1}$ (D) $e^2 + e + 1$

Ans. D

Sol. $f(1) = e$

$$\frac{dy}{dx} = \frac{1}{x} + e^x$$

$$\left.\frac{dy}{dx}\right|_{(1,e)} = 1 + e = m_T$$

$$m_n = \frac{-1}{1+e}$$

equation of normal $\quad y - e = -\frac{1}{1+e}(x-1)$

for x intercept $y = 0$

$\Rightarrow x = e^2 + e + 1$

30. Which one of the following statement is correct for function f(x)?

(A) f has maxima but no minima (B) f has minima but no maxima

(C) f has neither maxima nor minima (D) f has both maxima and minima

फलन f(x) के लिए निम्न में से कौनसे कथन सत्य है?

(A) f का उच्चिष्ठ है लेकिन निम्निष्ठ नहीं (B) f का निम्निष्ठ है लेकिन उच्चिष्ठ नहीं

(C) f का न तो उच्चिष्ठ है ना ही निम्निष्ठ है (D) f के दोनों उच्चिष्ट तथा निम्निष्ठ है

Ans. C

Sol. $f'(x) = \frac{1}{x} + e^x$

for $x > 0 \Rightarrow f'(x) > 0$

f is neither maxima nor minima.

MATCH THE COLUMN TYPE QUESTION(S)

31. Match the following column :

Column-I (Differential equation)	**Column-II (Integral curves)**
(A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$	(P) $y = A_1x^2 + A_2x + A_3$
(B) $(2x - 10y^3)\frac{dy}{dx} + y = 0$	(Q) $x^2y^2 + 1 = cy$
(C) $\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right) - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$	(R) $(x + 1)(1 - y) = cy$
(D) $(x^2y^2 - 1)dy + 2xy^3\,dx = 0$	(S) $x = A_1y^2 + A_2y + A_3$
	(T) $xy^2 = 2y^5 + c$

निम्न कॉलम को सुमेलित कीजिए :

Column-I (अवकल समीकरण**)**	**Column-II (**समाकल वक्र**)**
(A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$	(P) $y = A_1x^2 + A_2x + A_3$
(B) $(2x - 10y^3)\frac{dy}{dx} + y = 0$	(Q) $x^2y^2 + 1 = cy$
(C) $\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right) - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$	(R) $(x + 1)(1 - y) = cy$
(D) $(x^2y^2 - 1)dy + 2xy^3\,dx = 0$	(S) $x = A_1y^2 + A_2y + A_3$
	(T) $xy^2 = 2y^5 + c$

Ans. A – R, B – T, C – S, D – Q

Sol. (A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{y(1-y)}{1+x}$$

$$\Rightarrow \int \frac{dy}{y(1-y)} = \int \frac{dx}{1+x}$$

$$\Rightarrow \int\left(\frac{1}{y} - \frac{1}{(y-1)}\right)dy = \int \frac{dx}{1+x}$$

$\Rightarrow \ell ny - \ell n(y-1) = \ell n(1+x) - \ell nC$

$\Rightarrow yc = (1+x)(1-y)$

(B) $\left(2x - 10y^3\right)\frac{dy}{dx} + y = 0$

$\Rightarrow \frac{dx}{dy} = \frac{2x - 10y^3}{-y} \Rightarrow \frac{dx}{dy} + \frac{2x}{y} = 10y^2$

I.F. $= e^{\int \frac{2}{y} dy} = y^2$

$x.y^2 = \int y^2 \left(10y^2\right) dy$

$\Rightarrow xy^2 = 2y^5 + c$

(C) $\frac{dy}{dx}.\frac{d^3y}{dx^3} - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$

Let $\frac{dy}{dx} = v$ ...(1)

$\Rightarrow v \cdot v'' - 3(v')^2 = 0$

$\Rightarrow v.\frac{d^2v}{dx^2} - 3\left(\frac{dv}{dx}\right)^2 = 0$ ...(2) (v is variable of "x")

Now let $u(v) = \frac{dv}{dx}$ ...(3) [u is variable of v]

$u'(v)\frac{dv}{dx} = \frac{d^2v}{dx^2}$

$\Rightarrow u'(v)u(v) = \frac{d^2v}{dx^2}$ $\left[u'(v) = \frac{du}{dv}\right]$

put in equation (2)

$\Rightarrow v \cdot u' \cdot u - 3u^2 = 0$ or $u(v) = 0$

$\Rightarrow v\frac{du}{dv} - 3u = 0$

$$\Rightarrow \frac{du}{u} - 3\frac{dv}{v} = 0$$

$\Rightarrow u = cv^3$ or $u(v) = 0$

from equation (3)

$\frac{dv}{dx} = c_1 v^3$ or $\frac{dv}{dx} = 0$

$\Rightarrow \int \frac{dv}{v^3} = \int c_1 dx$ or $v = c_1$

$$\Rightarrow -\frac{1}{2v^2} = c_1 x - c_2$$

$$\Rightarrow v = \pm \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{c_2 - c_1 x}}$$ or $v = c_1$

Now $v = c_1$ or $v = \pm \frac{1}{\sqrt{2}} \frac{1}{\sqrt{c_2 - c_1 x}}$

using equation (1)

$\frac{dy}{dx} = c_1$ or $\frac{dy}{dx} = \pm \frac{1}{\sqrt{2}} \frac{1}{\sqrt{c_2 - c_1 x}}$

$y = c_1 x + c_2$ or $y = \pm \frac{\sqrt{2}\sqrt{c_2 - c_1 x}}{c_1} + c_3$

$y = c_1 x + c_2$ or $(y - c_3)^2 = \frac{2(c_2 - c_1 x)}{c_1^2}$

$\Rightarrow y = c_1 x + c_2$ or $c_1^2 y^2 - 2c_1^2 c_3 y + c_3^2 c_1^2 = 2c_2 - c_1 x$

$$x = \frac{-c_1 y^2 + 2c_1 c_3 y - c_3^2 c_1 + \frac{c_2}{c_1}}{2}$$

$$x = Ay^2 + By + C$$

M-2

$$y'y'''-3(y'')^2=0$$

$$\Rightarrow \frac{y'y'''-(y'')^2}{(y'')^2}=2 \qquad [\text{If } y''\neq 0]$$

$$\Rightarrow \frac{d}{dx}\left(\frac{-y'}{y''}\right)=2$$

$$\Rightarrow \int d\left(\frac{y'}{y''}\right)=\int -2dx$$

$$\Rightarrow \frac{y'}{y''}=-2x+c_1 \qquad \text{Let } y'=v$$

$$\text{then } \frac{v}{v'}=-2x+c_1$$

$$\Rightarrow \int \frac{dv}{v}=\int \frac{dx}{c_1-2x}$$

$$\Rightarrow \ell nv=-\frac{1}{2}\ell n(c_1-2x)+\ell nc_2$$

$$\Rightarrow v=\frac{c_2}{\sqrt{c_1-2x}} \Rightarrow \frac{dy}{dx}=\frac{c_2}{\sqrt{c_1-2x}}$$

$$\Rightarrow \int dy=c_2\int \frac{dx}{\sqrt{c_1-2x}}$$

$$\Rightarrow y=-c_2\sqrt{2}\sqrt{c_2-2x}+c_3$$

$$\Rightarrow (y-c_3)^2=2c_2^2(c_2-2x)$$

$$\Rightarrow y^2-2c_3y+c_3^2=2c_2^3-4c_2^2x$$

$$\Rightarrow 4c_2^2x=-y^2+2c_3y+2c_2^3-c_3^2$$

$$\Rightarrow k_1x=-y^2+k_2y+k_3$$

$$\Rightarrow x=Ay^2+By+C$$

or $y'' = 0 \Rightarrow y = mx + c$

(D) $\left(x^2y^2 - 1\right)dy + 2xy^3dx = 0$

$\Rightarrow x^2y^2dy + 2xy^3dx = dy$

$\Rightarrow x^2dy + 2xydx = \frac{dy}{y^2}$

$\Rightarrow \int d\left(x^2y\right) = \int \frac{dy}{y^2} + C$

$\Rightarrow x^2y = \frac{y^{-1}}{-1} + C$

$\Rightarrow x^2y^2 = -1 + cy$

$\Rightarrow 1 + x^2y^2 = cy$

32. Match the statements/expressions In Column I with the open intervals In Column II : **[JEE-2009]**

Column–I	**Column –II**
(A) Interval contained in the domain of definition of non-zero solutions of the differential equation $(x - 3)^2 y' + y = 0$	(P) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(B) Interval containing the value of the integral $\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$	(Q) $\left(0, \frac{\pi}{2}\right)$
	(R) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$
(C) Interval in which at least one of the points of local maximum of $\cos^2 x + \sin x$ lies	(S) $\left(0, \frac{\pi}{8}\right)$
(D) Interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing	(T) $(-\pi, \pi)$

कॉलम–I में दिए गये प्रकथन/व्यंजको का कॉलम–II में दिए गए विवृत अंतरालों से सुमेल करें : **[JEE-2009]**

Column–I	**Column –II**
(A) अवकल समीकरण $(x - 3)^2 y' + y = 0$ के अशून्य हलों के परिभाषा प्रान्त में समाहित अन्तराल	(P) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) अन्तराल जिसमें समाकल $\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$ (Q) $\left(0, \frac{\pi}{2}\right)$

का मान अन्तर्विष्ट है

(C) अन्तराल जिसमें कम से कम एक ऐसा बिन्दु जिस पर (R) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$

$\cos^2 x + \sin x$ स्थानीय अधिकतम हो। (S) $\left(0, \frac{\pi}{8}\right)$

(D) अन्तराल जिसमें $\tan^{-1}(\sin x + \cos x)$ वर्धमान है (T) $(-\pi, \pi)$

Ans. (A)P,Q,S (B) P,T (C) P,Q,R,T (D)S

Sol. (A) $\frac{dy}{dx} = -\frac{y}{(x-3)^2} \Rightarrow \ln y = \frac{1}{x-3} + c$

$\Rightarrow y = e^{\frac{1}{x-3}+c}$, $x \neq 3$.

(B) $I = \int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)\,dx$

Applying $x \to 6 - x$

$I = \int_1^5 (5-x)(4-x)(3-x)(2-x)(1-x)\,dx = -I$

$\Rightarrow I = 0$.

(C) $f(x) = \cos^2 x + \sin x$

$f'(x) = -2\cos x \sin x + \cos x$

$\Rightarrow \cos x(-2\sin x + 1) = 0$

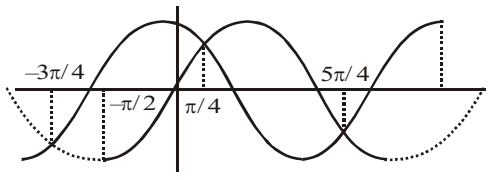
$\cos x = 0$ or $\sin x = \frac{1}{2}$

sign of $f'(x)$ changes from –ve to +ve while $f(x)$ passes through $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

(D) $f(x) = \tan^{-1}(\sin x + \cos x)$

$f(x) = \frac{\cos x - \sin x}{1 + (\sin x + \cos)^2} > 0$

$x \in (-3\pi/4, \pi/4)$



33. Match the statements/expressions given in Column I with the values given in Column II : **[JEE-2009]**

Column–I	**Column –II**
(A) The number of solutions of the equation $xe^{\sin x} - \cos x = 0$ in the interval $\left(0, \frac{\pi}{2}\right)$	(P) 1
(B) Value(s) of k for which the planes $kx + 4y + z = 0$, $4x + ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line	(Q) 2
(C) Value(s) of k for which $\|x - 1\| + \|x - 2\| + \|x + 1\| + \|x + 2\| = 4k$ has integer solution(s)	(R) 3
(D) If $y' = y + 1$ and $y(0) = 1$, then value(s) of $y(\ln 2)$	(S) 4
	(T) 5

कॉलम–I में दिए गये वक्तव्यों / व्यंजको का कॉलम–II में दिये गये मानों से सुमेल कीजिए : **[JEE-2009]**

Column–I	**Column –II**
(A) अन्तराल $\left(0, \frac{\pi}{2}\right)$ में समीकरण $xe^{\sin x} - \cos x = 0$ के हलों की संख्या	(P) 1
(B) k के मान (मानों) जिनके लिये समतल $kx + 4y + z = 0$, $4x + ky + 2z = 0$ तथा $2x + 2y + z = 0$ एक सरल रेखा में प्रतिच्छेद करते है (हैं)	(Q) 2
(C) k के मान (मानों) जिनके लिये $\|x - 1\| + \|x - 2\| + \|x + 1\| + \|x + 2\| = 4k$ का (के) हल पूर्णांक है (हैं)	(R) 3
(D) यदि $y' = y + 1$ तथा $y(0) = 1$, तब $y(\ln 2)$ का मान (मानों)	(S) 4
	(T) 5

Ans. (A) P; (B) Q,S ; (C) Q,R,S,T ; (D)R

Sol. (A) $x.e^{\sin x} - \cos x = 0$

$\Rightarrow e^{\sin x} = \frac{\cos x}{x}$

Let $f(x) = \frac{\cos x}{x}, g(x) = e^{\sin x}$

According the graph

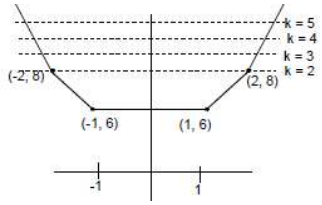
Number of solution is 1.

(B) $\begin{vmatrix} k & 4 & 1 \\ 4 & k & 2 \\ 2 & 2 & 1 \end{vmatrix} = 0$

$\Rightarrow k = 2, 4$

(C) $|x + 1| + |x - 2| + |x + 1| + |x + 2| = 4k$

graph will be



So, solution is k = 2,3,4,5

(D) $\frac{dy}{dx} = y + 1$

$$\int \frac{dy}{y+1} = \int dx$$

$\ln (y + 1) = x + C$

$(x + c)$

$y + 1 = e^{(x+c)}$

$y = c^{x+c} - 1$

$x = 0, y = 1$

$1 = e^{c} - 1$

$e^{c} = 2$

$C = \ln 2$

$y = e^{x + \ln 2} - 1$

for $x = \ln 2$

$e^{2\ln 2} - 1$

$y = 3.$